Moodify – Product Requirements Document (2-Day MVP)

Tagline: Feel it. Express it. Play it.

Version 1.0 · Scope: 2-day MVP

1. Product Overview

Moodify is a mobile app where a user describes how they feel in natural language (English, Hindi or Hinglish). The app detects the probable mood, asks the user to confirm it, and recommends Hindi and English songs that match. It learns from likes and dislikes and keeps a private mood history.

2. Problem and Goal

Problem: Choosing music for a mood takes effort, and generic mood playlists ignore language and personal taste.

Goal: Let a user go from "how I feel" to a fitting, personalised song list in under 10 seconds.

3. Target Users

- Students and young professionals who listen to Hindi and English music.
- People who want quick, low-effort music selection based on how they feel.

4. MVP Scope

In scope

1. User authentication and profile
2. Text mood input (English, Hindi, Hinglish)
3. LLM-based emotion detection with keyword fallback
4. Mood confirmation and correction
5. Song recommendation engine (mood, language, feedback)
6. Hindi and English song categories
7. Playback via external link (Spotify or YouTube)
8. Like, dislike and skip feedback
9. Mood history dashboard

10. Delete mood entries (privacy)

Out of scope (post-MVP)

Voice input, web version, Discover tab, collaborative filtering, in-app full streaming, Azure sentiment score, mood analytics beyond weekly counts. Stretch goal: 3-stage Emotional Journey playlist.

5. User Stories

ID	As a user, I want to...	So that...
US-01	sign up and log in	my data is saved
US-02	choose my languages and genres	songs match my taste
US-03	type how I feel in my own words	I don't pick from menus
US-04	confirm or edit the detected mood	wrong guesses are corrected
US-05	get a ranked song list with a reason	I trust the suggestions
US-06	open a song in Spotify or YouTube	I can listen instantly
US-07	like or dislike songs	future picks improve
US-08	see my mood history	I understand my patterns
US-09	delete my mood entries	my data stays private

6. Functional Requirements

ID	Requirement	Priority
FR-01	Register and log in via Supabase Auth	Must
FR-02	Onboarding saves preferred languages and genres	Must
FR-03	Accept free-text mood input in English, Hindi (Devanagari) and Hinglish	Must
FR-04	Return emotion, energy, intensity, context and target moods as JSON	Must
FR-05	Fall back to keyword rules if the LLM fails or times out	Must

ID	Requirement	Priority
FR-06	Show detected mood and let the user confirm or change it	Must
FR-07	Rank songs using mood match, language match and feedback	Must
FR-08	Filter by language (Hindi, English or both)	Must
FR-09	Show a one-line reason for each recommendation	Should
FR-10	Open song links externally	Must
FR-11	Record like, dislike and skip; exclude disliked songs	Must
FR-12	Show mood history list and weekly counts	Should
FR-13	Delete one or all mood entries	Should
FR-14	Provide fallback songs when external APIs fail	Must
FR-15	Emotional Journey playlist (acknowledge, reset, recharge)	Stretch

7. Non-Functional Requirements

- **Performance:** mood analysis returns within 3 seconds under normal conditions.
- **Security:** API keys stay on the backend; the app never calls the LLM directly; endpoints verify the Supabase JWT.
- **Privacy:** mood entries are private per user and can be deleted. The app presents moods as self-reported and AI-estimated, never as a mental-health diagnosis.
- **Reliability:** graceful handling of LLM or network failure with fallback songs.
- **Usability:** mobile-first, clear loading and empty states.
- **Accuracy:** test emotion detection on a labelled set of about 50 sentences per language, including Hinglish.

8. Mood Detection Design

Input: "aaj bahut thak gaya hu, kuch relax wala sunna hai"

Output:

```
{
  "emotion": "exhaustion",
  "energy": "low",
  "intensity": "high",
  "context": "work_pressure",
  "target_moods": ["relaxed", "stressed"]
}
```

Supported mood categories: Happy, Sad, Angry, Stressed, Relaxed, Motivated, Romantic, Nostalgic.

Language note: the prompt must state that input may be English, Hindi or Hinglish. Azure sentiment is optional and deferred; Romanised Hinglish needs separate testing.

9. Recommendation Logic

`Score(s) = w1·MoodMatch + w2·LanguageMatch + w3·FeedbackBoost`, then remove songs the user disliked.

- Songs come from a curated seed catalog of about 120 songs (60 Hindi, 60 English), each tagged with language, genre, moods, energy and link.
- Initial weights: mood 0.5, language 0.3, feedback 0.2. Tune after testing.

10. Architecture and Tech Stack

Layer	Choice
Mobile	React Native + Expo (expo-router)
Backend	Node.js + Express
Database and auth	Supabase (Postgres + Auth)
AI	LLM structured-output call + keyword fallback
Playback	External links (Spotify or YouTube)
Deployment	Render or Railway (backend), Expo Go or EAS (app)

11. Data Model

- **profiles:** user_id, name, preferred_languages[], genres[]

- **songs:** id, title, artist, language, genre, moods[], energy, url
- **mood_entries:** id, user_id, input_text, emotion, energy, intensity, confirmed_mood, created_at
- **song_feedback:** user_id, song_id, type (like, dislike, skip)

12. API Endpoints

Method	Endpoint	Purpose
POST	/api/mood/analyze	Text in, emotion JSON out
POST	/api/mood/confirm	Save confirmed mood, return songs
GET	/api/mood/history	Data for history and charts
DELETE	/api/mood/:id	Delete a mood entry
GET	/api/recommendations	Songs by mood and language
POST	/api/feedback	Like, dislike, skip
GET/PUT	/api/preferences	Languages and genres

13. Screens

1. Onboarding: login, sign up, language and genre selection
2. Home: greeting, quick mood chips, mood input, recent entries
3. Mood AI: text input, detected mood, confirm or edit
4. Recommendations: song cards, reason, like, dislike, open link
5. My Music: liked songs and history
6. My Mood: weekly counts, top genre, delete entries
7. Profile: preferences and settings

14. Timeline

Day 1 (backend, AI, data): setup and schema, auth and preferences, song seed file, emotion service with Hinglish tests, recommendation service and feedback API, mobile auth and onboarding.

Day 2 (UI and polish): Mood AI screen, recommendation list, Home and My Music, history chart, error handling and privacy, end-to-end testing, deployment and demo.

15. Success Metrics

- Emotion detection accuracy of at least 80 percent on the labelled test set.
- Mood-to-songs flow completes in under 10 seconds.
- At least 70 percent of confirmations accept the detected mood without edits.
- Zero crashes on API failure (fallback songs shown).

16. Risks and Mitigations

Risk	Mitigation
Hinglish detection is unreliable	Test early; use LLM prompt with examples; keyword fallback
Spotify audio-features endpoints may be restricted	Self-tagged seed catalog instead
Broken song links	Verify all links when building the seed file
Schedule overrun	Cut the chart first, then the Emotional Journey stretch goal
LLM latency or failure	Timeout plus fallback rules and default songs

17. Open Questions

- Final app name and branding
- Spotify vs YouTube as the default link target
- Whether to add Azure sentiment after the MVP